

Ideation Phase

Empathize & Discover

Date	13 July 2026
Team ID	SWTID-2026-7108
Project Name	BookStore
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures deeply rooted knowledge about a user's behaviors and attitudes. It serves as an essential tool to help engineering and design teams step outside of their own technical assumptions and better understand their target users. Creating an effective, user-centered digital system requires shifting our collective perspective to focus entirely on the true problem and the actual person experiencing it. By breaking down the target audience's external environment and internal thoughts, the team can map out real-world goals and challenges.

1. Detailed Empathy Map Canvas: The Everyday Reader & Gift Buyer

Below is the comprehensive operational mapping of our target user's everyday reality when attempting to browse, read reviews, find recommendations, and buy or reserve books online.

2. Core Synthesis & Next Action Items

By analyzing this Empathy Map Canvas, the development team can bridge the gap between abstract user feelings and hard system requirements:

1. Prioritizing Search Speed over Fancy Graphics: Because users feel frustrated spending 30 minutes scrolling[cite: 3], we must utilize MongoDB indexing to make multi-genre search and autocomplete instant.
2. True Mobile-First Coding: Since the user actively sees and hears about broken mobile sites while traveling[cite: 3], the React UI layout will be built using a mobile-first approach, testing touch target sizes and media queries for tablets and smartphones from day one[cite: 3].
3. Frictionless Checking & Reserving: To convert the pain of cart abandonment into a gain[cite: 3], the Node.js/Express.js backend will handle transactions and reservations smoothly in a single endpoint click.

Expanded Empathy Map Data Matrix (Tabular Format)

To extract clear architectural requirements for our MERN-stack platform, the unstructured empathy data is organized below into core, actionable pillars.

Dimension	User Experience Characteristics & Observations	Root Causes / System Triggers	Feature Solution Blueprint (MERN Stack)
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Think & Feel	- Desires an easy and exciting literary exploration process. - Values trusted community evaluations over generic ads. - Anxious about spending free time on clunky navigation.	- Flooded by excessive marketing data across the web. - Skeptical of standard automated sales tactics. - High-stress routines demand quick digital tools.	- Integrated User Reviews & Ratings module. - Clean, distraction-free UI layout. - High-performance React state management for zero-latency page transit.
Hear	- Peers discussing a desire for quick book reservations. - Digital trends emphasizing fast, responsive tech stack ecosystems. - Warnings about slow, frustrating web checkouts.	- Word-of-mouth recommendations driving traffic. - Rising consumer performance expectations across all e-commerce fields.	- Express.js and Node.js backend infrastructure designed for swift throughput. - Instant "Reserve & Collect" checkout option.

Dimension	User Experience Characteristics & Observations	Root Causes / System Triggers	Feature Solution Blueprint (MERN Stack)
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See	Chaotic online catalogs lack clear genre categorization.	- Outdated legacy database architectures. - Fixed-width web pages not designed	MongoDB flexible document structuring for advanced multigenre tagging.
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	- Content-heavy web designs that display poorly on mobile screens. - Viral book trends popping up on digital channels.	for fluid layout responsiveness.	Fluid CSS/Tailwind framework ensuring perfection on desktop, tablet, and mobile[cite: 3].
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Say & Do [cite: 3]	- Openly shares reviews but drops out when processes take too long[cite: 3]. - Moves across computers, tablets, and smartphones throughout the day[cite: 3].	- High interest in sharing knowledge combined with very low tolerance for bad UX[cite: 3]. - Dynamic, multi-device daily lifestyles[cite: 3].	1-Click action items for buying or reserving books[cite: 3]. - Cloud-synchronized MongoDB profiles for cross-device shopping persistence[cite: 3].
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Dimension	User Experience Characteristics & Observations	Root Causes / System Triggers	Feature Solution Blueprint (MERN Stack)
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Pain [cite: 3]	• Losing tracked items when changing devices[cite: 3]. • Missing out on timeless classics or new releases due to stock gaps[cite: 3]. • Wasting 30 minutes scrolling without discovery[cite: 3].	• Disconnected user accounts. • Slow database query setups lacking indexing. • Poor layout formatting[cite: 3].	• Autocomplete smart search engine powered by Node.js[cite: 3]. • Immediate structural error warnings and smooth, validated checkout fields.
Gain [cite: 3]	• Finding matches that fit accurate personal tastes in seconds[cite: 3]. • Confidently making successful gift purchases[cite: 3]. • Accessing the library on a phone anywhere[cite: 3].	• User feels smart, heard, and deeply efficient. • Reading journeys feel supported, fun, and easy[cite: 3].	• Custom Personalized Recommendation algorithms. • Highly responsive interface ensuring a smooth digital journey[cite: 3].